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Original Research Article

Behavioral public policy: past, present, & future

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Abstract

Behavioral public policy (BPP) applies behavioral insights to aid policy-making and implementation. Emerging from the interest in nudges in the late 2000s, BPP has matured over the last decade and a half. It has produced new concepts and generated much empirical data, helping researchers and policy-makers understand and explain human behavior with a multidisciplinary lens. Even though BPP is now an established field of endeavor, it has now reached the point when it is important to assess how likely it is to grow as a field. To help understand the dynamics of knowledge innovation and utilization, this article discusses the past, present, and future of behavioral public policy. We draw on the substantive achievements of nudge and BPP, the engagement of policy-makers in nudge units and behavioral policies, the growth of empiricism via experiments, and an emergent ethical concern illustrated by new tools, such as boost and nudge+. Responding to the critical dialog around nudge, we highlight the increasingly prominent role that agency and reasonableness are likely to play in future initiatives. We outline benefits that newer methods, such as computational social sciences, may have on the field. BPP can make sense of heterogeneity while not losing its ability to design simple, low-cost population-based interventions.

Keywords: behavioral public policy; nudge; nudge+; agency; reasonableness; computational social science

Academic fields typically evolve over decades, if not centuries, as disciplines mature and lay down foundations. Economic science is a good example. Despite its roots in the 18th century, economics only fully emerged as an independent and mature field of study toward the end of the 19th century. This independence, coupled with the marginalist revolution of the 1870s (Blaug, 1962), led to the creation of research chairs at leading universities and the founding of academic journals and professional societies dedicated to the growth of the field. For much of this period, there was an absence of a dominant school of thought. Various economic schools emerged in the early 20th century, often tied to national identities (such as the Austrian School). It was not until the 1950s and 1960s that neoclassical economics became dominant, characterized by "formal treatments of rational, or optimising, economic agents joined together in an abstractly conceived free-market, general equilibrium world" (Morgan, 2001, p. 7). Other disciplines within the social sciences have also followed a similarly lengthy path into maturity, such as political science.

An exception to this pattern of development is the field of behavioral science, especially when applied to public policy, today known as behavioral public policy (BPP) (Oliver, 2013). With its historical roots in (behavioral) economics and psychology, BPP rose to prominence remarkably quickly, mainly over the past 15 years or so. There are several reasons for its rapid emergence. The applied nature of the field and public and political support for behavioral regulation came at an opportune economic and political

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moment. The aftermath of the 2008 financial crisis, alongside growing distrust and apathy toward traditional economic approaches and the low cost of delivering behaviorally informed policies (Mills & Whittle, 2025), created the conditions for a less directive approach to public policy. Behavioral scientists borrowed heavily from advances in other disciplines, fostering an interdisciplinary research agenda. As a result, BPP is an established field of study using "insights from behavioural sciences, broadly defined, to shape and inform public policy" (Banerjee et al., 2024, p. 633). It is represented by its eponymous journal and an association of international academics who gather at an annual conference, which contributes to a favorable environment for its continued advancement. BPP is supported by related developments in applied economics and behavioral sciences, such as the US-based Behavioral Science & Policy Association (BSPA) and the European-focused Society for the Advancement of Behavioral Economics (SABE). Cognate journals, such as *Behavioral Science & Policy* as well as *Brain and Behavioral Sciences*, networks around the UK nudge unit, the Behavioural Insights Team (BIT), and its equivalents in other jurisdictions, along with related developments in adjacent fields, such as behavioral public administration (Grimmelikhuijsen et al., 2017), sustain this interdisciplinary research domain.

Looking back at the development of BPP, it is not controversial to say that the book *Nudge* (Thaler & Sunstein, 2008) was the major catalyst. Leading proponents and scholars of BPP would unanimously agree that this book propelled a behavioral age of regulation and ensured that the science and its applications entered the mainstream (John, 2018). This "trade book" generated worldwide interest, helped by its accessible language and gentle humor. It popularized the nudge tag and stimulated considerable interest by policy-makers in behaviorally and psychologically informed interventions. With behavioral units (erstwhile "nudge units") appearing in governments across the world (Naru, 2024), a track record of randomized controlled trials showing the potential for scaling up (List, 2022), and, in general, favorable reporting in old and new forms of media (John, 2018), policy-makers and policy-orientated academics have not looked back. Today's behavioral interventions are routine occurrences in public agencies across most fields of public policy, ranging from tax compliance (Antinyan & Asatryan, 2024) to environmental sustainability (Banerjee et al., 2023; for an overview, see Banerjee & John, 2023b). Of course, social marketing scholars and psychologists, especially in health, had been using and testing behavioral interventions for decades, but nothing compares with the scale and visibility of the nudge revolution that emerged since 2008.

As some agenda-setting models indicate (Downs, 1972), what goes up can come down, especially as long-term change may be harder to achieve than the early demonstration projects seemed to promise, and attention moves on to other more pressing topics. Nudge has attracted a lot of criticism from scholars who complain about its overstated effects and underlying paternalism (Baldwin, 2014; Bovens, 2009; Goodwin, 2012; Hansen & Jespersen, 2013; Sugden, 2009). The policy environment has now changed with strong state intervention getting more traction as many governments retreat from using market instruments, such as contracting, and right-wing administrations introduce top-down policies, such as over migration. Times of crisis, such as the coronavirus disease 2019 (COVID-19) pandemic, prompted a move from using soft toward hard policy instruments (Banerjee et al., 2021). Moreover, populist politicians might be suspicious of policy proposals emerging from the academy, such as the Ivy League universities. Even though BPP can work the slipstream, away from the public glare in the world of agencies and implementation, as happened during the first Trump presidency (John, 2023), it can still be seen as technocratic and elite-driven, so a target for the populist right. With all these changes in train, now is a pivotal moment for BPP. Its scholars must swiftly tackle the challenges to avoid becoming mired in critiques that could question the field's value and impact.

Rather than slipping down public priorities, there is a good chance that nudge and BPP can develop and innovate. It can engage with current debates in policy-making and conform to changing patterns of institutional design more generally. Rather than being an attractive add-on, BPP can join the mainstream. It has the chance of becoming a more systemized and adaptive form of public intervention whereby BPP practitioners understand public policy as a system, thereby providing long-term value. There are signs that BPP is headed in the direction of more systems thinking and greater use of new tools of intervention—developments that we chart in this article.

The structure of the article is as follows. First, we trace the origins of nudge and behavioral economics, reaching back to the 1960s and even earlier, and then note the progression toward a more explicit policy focus that led to nudge. Second, we assess the present and emergent alternatives, which show the growth of the field. We review the common limitations of nudge and BPP, focusing particularly on the recent criticism that nudge fails to account for the broader system (Chater & Loewenstein, 2023; Connolly et al., 2024a).

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We argue that nudge researchers have always considered the system, but recent methodological and evaluative innovations now enable a more comprehensive, integrative approach (Almaatouq et al., 2024). Finally, we outline an ambitious agenda for the future that envisages BPP moving beyond the confines of its standard tools. We outline novel approaches that seek to address behavioral problems more comprehensively. These approaches shift away from the standard lens of rationality dominant in behavioral economics and instead embrace reasonableness as a guiding principle. They also expand agency-enhancing strategies that empower individuals and promote a more mature dialog between citizens and the state. This tripartite structure encapsulates the goal of this article, which is an integrated review of BPP as defined by its past, present, and future.

Past: the origins and development of BPP

For as long as there has been social science, there have been studies of human behavior and recommendations about how to change it. Economists, such as Adam Smith, wanted to understand the conditions that fostered more competitive behavior in markets and the social support for such a system. Later, economists broadened the scope of the discipline to encompass different kinds of behavior, such as theories of crime, whereby the willingness of criminals to engage in illegal activities is based on their assessment of the costs and benefits of doing so (Becker, 2000). Broadly, a large section of modern neoclassical economics was underpinned by the workhorse model of logical rationality with cost–benefit calculations at its core. This branch of economics assumed people to be rational, with consistent preferences. It entailed the optimization of subjective utility functions, creating predictable equilibria that individuals and societies were thought to gravitate.

Since the 1960s, social scientists have taken much more interest in explaining human behavior. The rise of psychology as a discipline over the 20th century played an important role in shaping the behavioral research agenda across the social sciences, extending into studies and evaluations of public policy. Important within psychology was the school of thought called behaviorism (Skinner, 1965), which assumes that the task of explanation rests mainly on understanding observed behavior. With such a lens, the behavioral perspective comes to define the project of social sciences as a whole. Even though behaviorism attracted many criticisms, it had a large influence outside psychology, such as in the study of political behavior, which is influenced by psychology and focuses on observable acts, such as voter turnout, party choice, and political participation despite its interest in political attitudes and cultures that shape those behaviors (Sanders, 2018).

The aim of explaining human behavior in economics and psychology became more prominent in the 1970s when a team of psychologists put forward an ambitious research agenda called "heuristics-and-biases" (HB), which challenged the traditional tenets of logical rationality. The main proposition of the HB program was that humans often display biases, deviating from rational behavior, and often behave irrationally. They deploy shortcuts (heuristics) in preference to cost–benefit calculations. Tversky & Kahneman (1974) outlined three critical heuristics in their seminal paper in *Science*, namely (a) availability, (b) representativeness, and (c) anchoring and framing effects, all of which helped explain judgment under uncertainty. Kahneman and Tversky's blow to logical rationality fueled a broader research agenda, elaborating more heuristics. These works become foundational for the new field of behavioral economics (e.g. Kahneman & Frederick, 2002; Thaler & Shefrin, 1981; Tversky & Kahneman, 1981). Prior to this period, economics did not use many insights from psychology in theory building and testing. Human cognition was encased within the rational model, which requires the brain to calculate costs and benefits and then act on those assessments. If the brain could count benefits and discounts, then the micro-foundations of these models were thought to be intact. But these assumptions were not fully tested until Tversky and Kahneman carried out their landmark experiments.

There were exceptions of course, such as the work of Simon (1963) and studies of the psychology of taxation (Lewis, 1982). As Oliver (2017) points out, economists, such as Allais (1953), had always doubted that human beings acting in response to rational incentives are able to choose the utility-maximizing path. Critics from outside economics suggested a more plausible account of human reasoning offered by psychology would better serve the explanatory and prescriptive goals of social science, such as when trying to understand the behavior of policy-makers, who use heuristics and past guides to action to produce public policies (Allison, 1971). The experiments of Tversky & Kahneman (1974, 1981) complemented these critiques by applying core concepts in psychology to many familiar

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choice situations, showing that anchoring and other biases influence decision-making rather than utility maximization.

Early work in behavioral economics was theoretical, sharing with Allais an interest in paradoxes, as illustrated by Thaler's regular feature in the *Journal of Economic Perspectives* from 1987 to 1990 (Thaler, 1992). Over time, researchers became more oriented to real-world problems and in finding out what measures could overcome these biases and help individuals take what are thought to be more rational decisions. Research on pensions, for example, adopted a model of time-inconsistent preferences to justify the behavioral device of the default, which could be inserted into a pension scheme when someone started employment to get them into the habit of saving. This proposal was first tested by private companies (Thaler & Benartzi, 2004), reported in *Nudge*, then rolled out into public policies worldwide (Halpern, 2016; John, 2019; Sanders et al., 2023a). Defaults became frequently discussed in policy circles, such as to mandate the donation of organs on death (Johnson & Goldstein, 2003), which was then tested alongside other nudges in a large-scale randomized controlled trial (Sallis et al., 2018). Other important studies were contained in Oliver's edited volume, *Behavioural Public Policy* (2013), such as Julian Le Grande's survey of citizen preferences for paying for plastic shopping bags. He showed that citizens liked bag pricing as it helped them get to where they wanted to go, that is, to be better citizens who do not generate harmful waste. A neoclassical approach would predict that people would prefer free bags. The early work of BIT was influential, such as on fines (Haynes et al., 2013), electoral registration (John et al., 2015), using social norms for tax compliance (Hallsworth, 2014; Hallsworth et al., 2017), and social influences on charitable giving (Sanders, 2017). With these studies and interventions, BPP emerged as a field.

Of course, this rapid expansion of warrantable knowledge was followed by a period of "normal science" when findings bed down and become more commonplace. As a result of the diffusion of practice and extensive testing, there is a solid basis of knowledge of what interventions work or do not, and by how much they do in terms of effect size, such as in education (Damgaard & Nielsen, 2018). The experimental research agenda expanded to other policy issues, such as the environment (Linden et al., 2021), consumer affairs/finance, energy, food, vaccinations, and health behaviors (especially during the COVID-19 pandemic). Field-specific meta-analyses show positive effects of these interventions, such as for tax compliance (Antinyan & Asatryan, 2019) and food waste (Zhang et al., 2023), whereas alternative meta-analyses of behavioral interventions to promote action on climate change yield more modest conclusions (Linden et al., 2021). We discuss meta-analyses in the next section.

The error correction approach to designing public policies did not reign supreme. It was challenged by another competing theory from psychology, ecological rationality, whose protagonists posited that heuristics could be helpful in individual judgment under uncertainty. Core to this agenda was the proposition that it would be misleading to associate heuristics systematically with biases. The work of Gigerenzer and colleagues (Kruglanski & Gigerenzer, 2011; Gigerenzer & Goldstein, 1996; Gigerenzer et al., 2000, 2016; Gigerenzer & Gaissmaier, 2011; Gigerenzer, 2024) showed how smart and simple heuristics could lead to reasonable (and rational) answers even if humans struggle to make rational deductions. These alternative theories of decision-making and debates around them have recently been summarized by Gigerenzer (2024) as the rationality wars.

Present: criticisms of nudge

Fifteen years have passed since the UK government began integrating nudges—and behavioral science more broadly—into day-to-day policy-making and implementation. But nudge was met with skepticism right from the outset. Following the initial trial period of nudge-based policies under the guidance of BIT, a review process, termed the "Sunset Clause" was conducted to assess their cost–effectiveness in governance. In its 2011 report, the UK House of Lords Science and Technology Select Committee acknowledged the appeal of nudges but questioned their long-term viability in addressing society's major challenges. The report emphasized:

> Non-regulatory measures used in isolation, including 'nudges,' are less likely to be effective. Effective policies often use a range of interventions. We concluded that it is important to consider the whole range of possible interventions when policy interventions are designed. We place particular emphasis on this conclusion because the evidence we received indicated that the Government's preference for

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> non-regulatory interventions has encouraged officials to exclude consideration of regulatory measures when thinking about behaviour change. (p. 5, 2011)

These concerns were reflected in academic criticism of nudge (e.g. Goodwin, 2012; Marteau et al., 2011), which drew attention to the advantages of conventional regulation. Other critiques of nudge appeared from within the behavioral sciences community (Chater & Loewenstein, 2023; Hallsworth, 2023a) in an attempt to diversify the toolkit that BPP promises to offer. Chater and Loewenstein (2023) argued that an overreliance on individual-focused interventions—the so-called i-frame—had diverted behavioral public policy away from more systemic and impactful approaches. This growing discontent with the nudging approach can be categorized into three main themes.

Semantical challenges

One of the earliest challenges of nudging—widely synonymous with behavioral public policy in its formative years—was the lack of conceptual clarity surrounding the term nudge. Defined by Thaler & Sunstein (2008), a nudge refers to an intervention that neither bans options nor makes them economically costly. Instead, it influenced decision-making by presenting choices differently, aiming to help individuals make decisions they would deem beneficial if they were acting rationally.

Central to this approach was the condition that a nudge could predictably alter behavior by bypassing cognitive shortcuts (heuristics)—the mental rules of thumb that often led to biases or deviations from rational behavior. This framework was rooted in the heuristics and biases paradigm, discussed earlier, which the nudging framework incorporated. Nudges, as originally defined, were guided by the political philosophy of libertarian paternalism: the idea of encouraging people toward better outcomes while preserving their freedom of choice.

These criteria were conceptually defensible to many semantic scholars, even though the governance style itself faced criticism. Early applications of nudging adhered to these boundaries, such as 401(k) default pension schemes (Thaler & Benartzi, 2004). These schemes automatically enrolled individuals to save a higher proportion of their income, adjusting contributions with annual salary increments to minimize the psychological aversion to perceived losses.

The scope of nudges gradually expanded beyond Thaler and Sunstein's original definition to include interventions with limited predictability in their outcomes. For instance, information campaigns were often classified as nudges, despite being broadly educative and failing to maintain an unchanged choice set—a key criterion for a nudge (Hansen & Jespersen, 2013). John (2013) went even further by arguing that "all tools of government are informational now" by having a nudge or informational component embedded within them. Over time, additional classifications of nudging emerged (see Banerjee & John (2023b) for a review), leading to what Haslam (2016) described as "concept creep" (see also "concept expansion" proposed by Sartori (1970)). Haslam uses this term to refer to the gradual expansion of definitions in psychology, resulting in the dilution of their original meaning. Nudging has suffered from conceptual expansion.

This conceptual broadening or stretching has made it increasingly difficult to contextualize and compare nudges meaningfully. Many meta-analyses, while necessary, treat all nudges as comparable despite their differing mechanisms of action, rendering these comparisons cumbersome to interpret (as discussed above). Responses to this conceptual impurity have varied. Some scholars, who have also proposed frameworks to classify and compare nudges in better ways, such as the policy cube (Banerjee, 2021; Oliver, 2018), have later called for abandoning such buzzwords altogether, advocating instead for use based on the underlying objectives and mechanisms of interventions (Oliver, forthcoming in this issue). However, this issue remains unresolved, and more definitions of nudges are increasingly proposed in the literature.

Ethical challenges

A recurring critique of nudging has been the lack of transparency in its design and delivery. One of the earliest and most widely cited criticisms is that nudges are effective because they "work in the dark" (Bovens, 2009). This perceived lack of transparency often renders nudges covert and exposes them to allegations of manipulation. Nudges, by definition, should be easy to opt out of. However, ethicists argue that they can differentially affect individuals' psychological capacities, making it difficult for some to recognize or resist them (Ivanković & Engelen, 2019).

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This debate on transparency in nudging has inspired several research themes. For instance, Baldwin (2014) categorized nudges into three types, based on the extent to which they allow recipients to recognize and respond to the nudge autonomously. Similarly, Hansen and Jespersen proposed a framework mapping nudges along two dimensions: transparency and degree of cognitive engagement, resulting in four distinct classes of nudges. Others (Bruns & Paunov, 2021; Michaelsen, 2024; Paunov et al., 2019) categorized different types of transparency, highlighting, often using experiments, how various forms of disclosure impact the effectiveness of nudging. Lades & Delaney (2019) further extended this discussion with their FORGOOD ethical framework, offering a comparative lens for evaluating different behavioral public policies, including nudges.

Another line of research, driven by an empirical focus on quantifying the effects of more transparency in nudging, examines the impact of disclosures after nudges have been delivered. For example, Loewenstein et al. (2015) found that informing people they are about to be nudged did not diminish the nudge's effectiveness. Similarly, Sunstein and colleagues (Reisch et al., 2021; Sunstein, 2016a, 2016b) provided cross-country survey evidence that nudging is widely supported, especially for interventions that promote reflective thinking (System 2 nudges). Taken together, these responses to critiques of nudging broadly compartmentalize into two forms: either making the nudge transparent prior to its delivery (the ex ante defence) or asking people afterwards how they feel under the influence of the nudge (the ex post defence). Both approaches, however, fail to deal effectively with the criticism that they do not make provision for people who do not want to be nudged at all.

Inspired by these treatments, Wilkinson (2013) put forward a third proposition by asking for people's consent before nudging them. Recent scholarship has put this to test by obtaining consent and violating it randomly in subsets of people. Gerver et al. (2024) show that while respecting consent does not reduce the effectiveness of nudges in eliciting donations, it significantly enhances individuals' self-reported well-being measured by their increased levels of happiness and support for the policy, as well as decreased resentment and regret against the nudge. Although evidence on the role of consent-seeking nudges remains novel and sparse, it does add a new form of managing ethical concerns leveled against nudging.

To summarize, some scholars call for more individual agency in behavioral public policy (Banerjee et al., 2024), while others advocate for reduced agency in favor of systemic interventions (Chater & Loewenstein, 2023). Between these extremes, proponents of nudging continue to navigate between effectiveness, transparency, and ethics in ways that can promote equality between persons as core to nudge.

Efficacy challenges

The third critique of nudging concerns its variability in effectiveness. While this criticism is less robust than others, for semantical reasons discussed earlier, it cannot be entirely dismissed. The challenge lies in the fact that many nudges fail to achieve the desired level of effectiveness as policy tools, raising questions about their scalability.

The first significant critique of nudging effectiveness came from the meta-analysis by Mertens et al. (2022). This analysis revealed substantial variation in the effectiveness of nudges across different domains and types. For example, decision-information nudges, such as those enhancing salience and visibility, yielded a moderate average effect size of $d = 0.34$. Decision-assistance nudges, including commitment devices and reminders, performed worse, with an average effect size of only $d = 0.28$. In contrast, decision structure nudges, such as defaults, positioning, or placements, exhibited a much larger effect size of $d = 0.54$.

Moreover, effectiveness varied significantly by domain. Nudging in the food sector showed much higher effectiveness ($d = 0.65$) compared to nudging in financial ($d = 0.34$), health ($d = 0.34$), or other decisions ($d = 0.31$). Responses by Maier et al. (2022) and Bakdash & Marusich (2022) criticized the methodology of Mertens et al., applying stricter corrections for truncated or file-drawer effects, which further reduced the reported effect sizes. While results have been more strongly corrected in Maier et al. (2022), few would want to assume the null with behavioral interventions, and most agree with Hallsworth (2022) that there is a lot of complexity and variation under the bonnet of behavioral interventions which vary between and within each domain of intervention.

While one interpretation of these findings might be to dismiss nudging as only weakly effective, a more constructive response is to explore the factors driving these differences. The mixed-effects model later presented by Mertens et al. (2022) highlights that study characteristics significantly influence nudge variability. This aligns with broader discussions in the field, such as those by Beshears & Kosowsky (2020),

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which emphasize the importance of understanding the mechanisms underlying nudging. Such understanding can help clarify the causal pathways through which nudges and other behavioral interventions operate, enabling better classification and customization of these tools (Banerjee & Veltri, 2024; Grüne-Yanoff, 2016). For example, there is also an effect across the different kinds of interventions relying on different mechanisms, such as reminders, commitment devices, norms, defaults, and visibility, all showing positive effects. However, while this need is widely recognized, it remains underexplored—leaving mechanistic diagnosis as a glaring oversight in the field.

Another significant limitation of nudging is the lack of trials with a temporal design. Many behavior changes require sustained intervention over time, yet most studies rely on one-shot experiments. This approach risks overlooking the non-persistence of nudge effects. For instance, DellaVigna and Linos (2020) report that treatment effects typically decline by 7–8 percentage points following the withdrawal of nudge policies. It should be noted that other tools of public policy could also suffer from this critique (such as a tax in the absence of which demand can theoretically increase again). Additionally, as nudges become more widespread and long-term, individuals may adapt to them, reducing their effectiveness.

Finally, scalability issues may also arise from other factors, as explored by Dellavigna & Linos (2020), who analyzed 126 RCTs in the United States and found a "voltage drop"—a consistent decline in nudge effectiveness in real-world settings. These findings underscore the importance of understanding and addressing the limitations of nudging in BPP so that policy-makers can learn how to enhance its efficacy as a policy tool.

The systems approach

The argument for the systems approach needs to bear in mind that nudge advocates always had the s-frame as their ultimate objective. The reason why individual nudges are so important is that policy-makers need to see a proof of concept if they were to authorize nudges, and that behavioral units need to make the case for more funding. A start-up into system nudges does not make sense in the competitive context of agencies and units within bureaucracies. Academics find the one-off nudge to be useful for the article or short paper where the contribution can be stated and various robustness tests performed, so they are happy to collude in the proof-of-concept approach, especially as system interventions are bound to take a long time. But the intention of any one-off trial is that the findings feed back into policy in a process of redesign. Once the proof of concept has been made, it is easier to propose the next step to influence complex environments. This is well known in evaluation research, in which BPP is often seen as a part, so familiar methods such as the staged or pragmatic RCTs are tailored to seeing how carrying out the intervention affects the take-up of a policy. Meta-analysis can be helpful in fulfilling long-term aims of policy-makers by offering understanding of the effects of interventions in a wide range of contexts and is designed to support scaling up (Al-Ubaydli et al., 2021; List, 2022).

Even though the i- and s-frame criticism was based on an oversimplification as behavioral studies already take into account the wider system (Sunstein, 2023), the critique did hit home (Andreas & Jabakhanji, 2023), and was a prompt for behavioral researchers to be more explicit about the context for their research (Connolly et al., 2024b; Herzog & Hertwig, 2025). This critique emerges in Hallsworth's manifesto (2023a) with his invocation to "see the system". It is likely that researchers will try to pay more attention to system effects in future studies.

It is less clear what a systems approach entails for both research and intervention designs. It is always easier to do the criticizing than to map out what s-frame interventions look like. The basic idea is that a nudge on its own is unlikely to reveal a prescription for policy design, as the conditions for a nudge being introduced are likely to be special, which is a commission by an external partner, so that the nudge will likely be working in a relatively artificial context. There is a need to commission several nudges to ensure these studies and demand effects are accounted for, and it is possible to test for a nudge in normal and business-as-usual environments. The systems approach could build on how trials and research knowledge can take an adaptive perspective, such as pragmatic trials. The use of techniques from qualitative or mixed methods research can be useful in promoting a systems approach. For example, Stoker & John (2009) developed the design experiment method whereby researchers and policy-makers continually tweak an intervention by feeding back the results from the evaluation into a new design, which is tested again. Successive redesigns ensure that all the kinks get ironed out and outcomes are thereby improved (Askew et al., 2010).

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Future: the need for an alternative lens for BPP

In this section, we gaze into our crystal ball to look at the future, which is naturally a risky endeavor. We outline four promising avenues of work that we believe BPP will gravitate toward over the next few years, which build on its strengths but also address, to some extent, the criticisms that have been mounted, as reviewed in the previous section.

Reasonableness

Behavioral economics and science have largely relied on the notion that humans are "irrational" when they make choices; hence, the need for a nudge to move people toward welfare-improving choices. This form of libertarian paternalism, however, runs the risk of belittling human beings and undermining their capacities. Labelling people and their actions as irrational is increasingly being challenged, with a growing call by practitioners to move away from it as it "may increase overconfidence and impede deeper inquiry" (Hallsworth, 2023b, p. 2). While delegitimizing irrationality might be necessary for the future of behavioral sciences, it is by no means sufficient, argues Hallsworth, because behavioral scientists need to become more cognizant of contexts and institutional processes, which can explain why humans behave the way they do in different settings. An alternative approach is to assume that people are and can be reasonable (Madsen et al., 2024). The concept of reasonableness has existed in psychology for a while, for example, it has been recently traced back to early works by Kahneman in creating a more coherent and time-consistent explanation of how humans can factor in conditions such as regret (Banerjee & Dold, 2024). However, it has not yet been fully developed. Reasonableness has the benefit that it treats individuals as targets of interventions, thereby highlighting the need to involve and engage them. By putting citizens at the heart of behavioral public policy (Banerjee & John, 2023a), reasonableness as an approach can build partnership and dialog with people and "acknowledge the systems they inhabit". It is here that approaches like the "'nudge plus' [have] the potential to develop a more cyclical process such that citizens are encouraged to feedback to policy makers about the way in which they would like choice environments that shape their behaviour to be structured" (Madsen et al., 2024, p. 585).

There are risks, nonetheless, of building participatory approaches into behavioral public policy, as has been indicated in previous research where demographic and attitudinal factors explained differential participation rates (Einstein et al., 2020; Hanson et al., 2025). Participation is often unequal across society, and there is a high degree of self-selection involved in traditional participatory approaches. People-centered approaches, such as the nudge+ toolkit, help prevent this self-selection by encouraging each individual to think alongside the nudge such that reflection is made proximate to the delivery of the nudge. While this mitigates this concern to some extent, we note that there might be differences that arise as different individuals have different intrinsic capabilities to reflect, which can widen the inequalities of nudges. The same also holds for boosting interventions, where compliance with the intervention is dependent on the inherent abilities of individuals. Both these toolkits therefore impose some constraints on individuals and do not work for the whole population. So a mix of strategies is therefore key. However, recent advancements in participatory methods suggest that traditional participatory approaches could be improved with newer and more inclusive methods of public engagement. For example, the participatory value evaluation (PVE) method developed by Mouter et al. (2019) provides a costless and easy way of engaging people inclusively in preference elicitation tasks and have been shown to be effective across a wide range of decisions, such as urban mobility investments (Mouter et al., 2021a), flood protection schemes (Mouter et al., 2021b), and healthcare decision-making (Boxebeld et al., 2024). By designing participatory environments more innovatively, many risks of self-selection can be minimized.

Agency

The trends to stress agency appears in the work on boosts (Herzog & Hertwig, 2025), nudge+ (John & Stoker, 2019; Banerjee & John, 2023a), and debiasing (Banerjee et al., 2024), but is the nature of these claims that not enough work is currently being done to improve agency. It could be made more normal rather than being the preserve of research programs. At the time of writing, most behavioral interventions do not stress agency, except in a limited sense of seeking consent at the start of the study and following basic ethical procedures, especially for research projects. The question is whether some agency is a consequence of the change of research orientation and could be scaled into most projects, especially

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ones needing long-term engagement of participants, such as s-frame projects. We think this is a realistic possibility, with reflective interventions and feedback becoming normal parts of nudge interventions. Nudge+ could become just the nudge of the future by being a common element in behavioral interventions. It is also important to understand the trade-offs that agency can have with effectiveness. While some scholars justify calls for more agency today on grounds of effectiveness (Banerjee et al., 2024), others call for the opposite, as individual interventions that preserve freedom of choice are seldom very effective (Chater & Loewenstein, 2023). The present-day nudging approach is somewhere in between. This trade-off has been implicitly assumed and discussed in the context of BPP many times, but there is no formalization of how such trade-offs can materialize. We believe BPP would benefit from such a contribution.

Computational social science

It is increasingly evident that human beings are inherently diverse, and behavioral interventions should reflect this diversity (Banerjee & Galizzi, 2024b). BPP has gained significant momentum through "what works" studies, which prioritize identifying the most effective interventions. While this approach was justified in the early days of BPP, particularly when research programs directly tied to public policy were required to demonstrate cost-effectiveness, it has inadvertently sidelined an equally important question of why interventions work in the way they do?

The early years of nudging provide a notable example. BIT was tasked with demonstrating returns exceeding its initial investment, effectively proving the economic value of nudging (Halpern, 2016). Over time, nudging-style interventions have outperformed many traditional governance and regulatory mechanisms (Benartzi et al., 2017). However, the field's preoccupation with evaluating effectiveness has often come at the expense of understanding mechanisms. This oversight has led to persistent criticisms of BPP, many of which we have discussed earlier in this article, particularly when interventions fail to scale or adapt to diverse contexts. A one-size-fits-all approach, where interventions are applied uniformly across regions and cultures, disregards the underlying heterogeneity of human behavior. Addressing this shortfall requires a paradigm shift in BPP research from "what works?" to "why does it work?"

Advances in computational social science (CSS), including machine learning and large-scale behavioral modeling, provide the tools to explore these mechanisms underlying intervention success (Athey & Imbens, 2016; Veltri, 2023). By embracing these methods, researchers can uncover the pluralism inherent in human behavior and develop more nuanced, tailored, and personalized behavioral interventions (Banerjee & Veltri, 2024). A deeper understanding of mechanisms will enable us to address heterogeneity in intervention uptake, ultimately enhancing the efficacy of BPP toolkits. Mechanism understanding allows for greater precision with this policy instrument, which can be gained with more attention to subgroups, thereby harnessing resources and avoiding unnecessary communications and possible reactance from other groups. But there is still room for the population-wide, light-touch interventions that characterizes the first phase of nudge. While computational social science can uncover demographic and attitudinal traits that explain treatment uptake, thereby enabling an understanding of "for whom does the intervention work?", sometimes this is limited in scope to get to "why does it work"? This is because these methods can be data-intensive and therefore might not always be feasible to implement. In such cases, and more generally, carefully thinking through mechanism design can also help with more robust tests of theory, aiding the use of such computational social science methods. For example, unlike conventional wisdom that suggests that lack of awareness causes vaccine hesitancy, Wilhelm et al. (2024) recently show in a sample of French healthcare workers that it is not awareness but situational reactance, which led to resistance to vaccination in a sample of French healthcare workers. Similarly, De La Rosa et al. (2021) show that psychological ownership increases people's willingness to claim government benefits by reducing people's general aversion to asking for assistance. While CSS methods can aid in understanding why interventions work, they are certainly no silver bullet, as they are limited and therefore must also be complemented by theoretical approaches to uncover mechanisms.

Lastly, one of the related benefits of these newer methods in computational social science is that they could also be used to detect differential treatment effects of behavioral public policies, which then help researchers and practitioners understand how BPP toolkits impact different groups of people and use this heterogeneity in better designing less distributionally adverse policies. The recent rise of synthetic participants is also an exciting development that complements this heterogeneity revolution, as policies

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can now be tested using non-human participants (Shrestha et al., 2025), which allows for a better understanding of differential treatment effects, therefore enabling a more cautionary delivery of newer behavioral policies.

Diversity and inclusion

Much of the early work on behavioral sciences for policy has focused on evaluating the impact of behavioral insights in encouraging welfare-improving behaviors of a limited part of the global population, the so-called Western, Educated, Industrialized, Rich, and Democratic (WEIRD) country participants. This has led to generalizability problems due to contextual differences between WEIRD and non-WEIRD countries, a distinction that has been criticized to be too naïve lately (Ghai, 2021). Today, there is an increasing effort to make BPP more inclusive, which is evident both in policy advice and academic research (Ghai & Banerjee, 2024) as well as the groundwork that is being carried out in the global south (Sanders et al., 2023b) to create a more systematic and holistic approach to tackling global challenges, such as climate change (Banerjee & Galizzi, 2024a).

Building this sample diversity is much needed as we embrace newer experimental methods to test and evaluate public policy interventions (Ghai & Banerjee, 2024). There are also distributional effects as some behavioral public policies can harm certain population segments, as has been discussed by Sunstein (2022) lately. For example, Hall & Hernandez (2024) show that many policy interventions developed and tested are often implicitly colorblind, thereby exacerbating systematic differences along race or ethnicity. Being oblivious to such distributional effects, or limiting studies to certain population segments, therefore raises ethical questions about the use of older (such as nudging) and newer (such as people-centered programs) forms of BPP.

We believe that BPP will increasingly embrace more comparative work with diverse population samples that allows academics and practitioners to understand the overall general consequences, including distributional effects, of implementing such policies and boosts the external validity and generalizability of BPP findings.

Conclusions

This review of the past, present, and future of BPP shows how fast this field has developed from its early beginnings in the 1970s to the upswing of the 2000s and then to the more settled pattern of today. In agenda-setting models, attention goes up and often comes down (Downs, 1972), but even the most cyclical of agenda-setting models says that attention sustains over time, even at a lesser rate. Recent agenda-setting models (Baumgartner & Jones, 2010; Jennings & John, 2010) suggest that a change in attention, either upwards or downward, tends to be a punctuation that continues over time. This sustained upswing has happened with BPP, hence the prediction of a steady state and BPP becoming business as usual for both agencies and academics.

BPP is now established and embedded both as an academic field as well as a toolkit that policy-makers know about and use regularly. The latter can either be done in dedicated nudge units or by researchers and policy-makers working together within existing governmental units, or by commercial organizations and businesses who are using it to improve customer experiences and/or profits, and who often work with public sector clients. We believe that this engagement with behavioral public policy will remain an important approach that policy-makers are likely to continue using. It will appear less in nudge units (which are also diversifying in their approaches), but more in businesses or in more general evaluation units, which will have a behavioral application and are increasingly setting up such teams. Examples of behavioral units in private firms abound today, such as the Boston Consultancy Group's Behavioral Change Lab, which uses insights from behavioral science, psychology, and neuroscience to inform and improve its clients' practices. Increasingly, banks and consulting firms are also setting up smaller units within their organizations that are using behavioral insights to inform their working policies.

Bedding down and business as normal has a lot of advantages for BPP. Its cumulative program suits an incremental strategy of many tests and continuous tweaking. The innovations charted in the section on the future provide a way for BPP to keep on offering new benefits, albeit within a normal science paradigm. The continued attention to BPP by academics and policy-makers lies in the realism of its

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research and policy agenda, which addresses some of the weaknesses of rational actor models. Its use of more intuitive and diverse behavioral models is an expression of the general maturation of social sciences to using a more considered behavioral model of human action, whilst not forgetting the important insights of rational-actor models, such as for collective action. The practical solutions that BPP offers to common public problems help explain its appeal and its utility for policy problems, as well as its contribution to academic debate.

We also expect behavioral public policy to continue as a lively area for scholarly work, more as a hybrid field rather than a fully-fledged discipline. This is mainly because BPP continues to draw heavily from concepts and methodologies used in more established disciplines, such as economics and political science. BPP scholars often pursue interdisciplinary research, nonetheless retaining their disciplinary focus in publication conventions and use of methods. The many critiques from academics working inside and outside BPP encourage a lively debate and a search for new tools that seek to address those critiques. Even with all this activity and research, it is difficult to foresee BPP developing into a new discipline, complete with its own set of theories and school of thought which is fully distinct from other mature fields of study within the social sciences.

As both an academic and policy endeavor, the move from shiny new to normal presents some disadvantages. Policy-makers might be less likely to commission nudges as exciting new things, so that the high-level authorizations of the past are less common. There is also the risk that, in a world where behavioral insights are becoming so commonplace, and where citizens are increasingly made cognizant of the behavioral applications being done to them, these insights to foster behavior change may not retain their efficacy. As we discussed, moving toward a reasonable citizenry, which is empowered to act, might be a better alternative to more top-down nudges. Such people-centered approaches are becoming more common in behavioral science applications to specific domains, such as in environmental policy (OECD, 2025), and there is reason to believe that this trend can be extended to other applications too. New methods and approaches improve the understanding of experiments, exploring heterogeneity and outlining mechanisms, which can help nudges become more efficacious and cost-effective.

Much of the early success of the nudge program was due to its convenience and ease of delivery, which led to its early adoption and deployment, and consequently cost-effectiveness (Benartzi et al., 2017). As an alternative, it is likely that this new program of people-centered BPP approaches has limitations in scaling, especially when thinking about conventional delivery forms of public policy, such as deliberation in groups or thinking in pairs, both of which have happened in physical settings and therefore have been costly to scale. However, with the growth of digital platforms and the possibility to use computational social science, it is likely that agency-enhancing interventions can also be delivered at scale. For example, most nudge+ prompts could be introduced as pop-ups that can be easily displayed in kiosks. Reflection prompts via mobile applications could also help policy-makers collect data on preferences of people, which then becomes the starting point for personalization of behavioral interventions, forgoing the need for policy-makers to assume what is truly beneficial for citizens "as judged by themselves."

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